

CS222: Computer Organization & Architecture

End Semester Exam, Spring, 2013

Full Marks: 80

Time: 3 Hours

Answer all questions.

1. Answer the following in brief.

- (a) What is memory mapped I/O?
- (b) Why is DMA an improvement over CPU programmed I/O?
- (c) When would DMA transfer be a poor choice?
- (d) Mention two advantages and disadvantages for using a single bus as a shared communication link between memory, processor and I/O devices. (2+2+2+4)

2. Consider the following piece of code:

```
int x = 0, y = 0; // The compiler puts x in r1 and y in r2.
int i; // The compiler put i in r3.
int A[4096]; // A is in memory at address 0x10000 ...
for (i=0; i<1024; i++) {
    x += A[i];
}
```

Assume that the system has a 8192-byte, 2-way set associative data cache with 16-byte sets (8-byte blocks). Assuming that the cache starts out empty, what is the series of data cache hits and misses for this snippet of code. Assume that ints are 32-bits. What is the total number of cache misses? (10)

3. Computer A has an overall CPI of 1.3 and can be run at a clock rate of 600MHz. Computer B has a CPI of 2.5 and can be run at a clock rate of 750 Mhz. We have a particular program we wish to run. When compiled for computer A, this program has exactly 100,000 instructions. How many instructions would the program need to have when compiled for Computer B, in order for the two computers to have exactly the same execution time for this program? (5)

4. Convert the C function below to ARM assembly language. Make sure that your assembly language code could be called from a standard C program (that is to say, make sure you follow the calling conventions).

```
unsigned int sum(unsigned int n) {
    if (n == 0) return 0;
    else return n + sum(n-1);
}
```

[Register convention for ARM, r0-r3 – Argument registers (not preserved), r4-r11 – Variable registers (preserved), sp – Stack pointer (preserved), lr – Link register (preserved), r12 – Scratch register for inter-procedure call (not preserved)] (10)

5. (a) Using the code below, list all of the dependence types (RAW, WAR, WAW). List the dependences (example INST-X to INST-Y) by writing in the instruction numbers involved with the dependence.

```
I0: A = B + C;
I1: C = A - B;
I2: D = A + C;
I3: A = B * C * D;
I4: C = F / D;
I5: F = A ^ G;
I6: G = F + D;
```

(5)

6. Describe a pipelined datapath needed to support only 'add rd, rs, rt' instruction. Your datapath should be designed with the assumption that only this type of instruction will be executed. (10)
7. What is virtual memory? Explain page faults and TLB. (3+4+4)
8. What is network and bisection bandwidth? Compute these for processor network connected in cube fashion. Assume there are $N = 2^n$ processors. (5+5)
9. Explain with example how various hazards are handled in dynamic scheduling with a scoreboard. (10)